

DJNZ Register, Label

Jump only if a certain condition is met

JZ $\rightarrow$ Z = Zero

JNZ $\rightarrow$ NZ = Non-zero

JZ label ; jump if A=0

if R5 contains the value = 0 ; put 55H in it.

MOV A, R5

JC $\rightarrow$ C = carry

JNC $\rightarrow$ NC = No carry

JB $\rightarrow$ Bit = 1

JNB $\rightarrow$ Bit = 0

DJNZ = Decrement and Jump if A $\neq$ 0

CJNE #
Later

{ LJMP (long jump) vs SJMP (short jump) } useful for CT

(05 + 0D = 12) # page 112

Hex	Decimal	F2 $\rightarrow$	15	2
		15	2	7
F	15			
10	16	F2 $\rightarrow$	<u>1111</u>	<u>0010</u>
11	17	+ 15	<u>0001</u>	<u>0101</u>
12 $\leftarrow$	18		<u>107</u>	<u>10000 0111</u>

Call Instruction

- ① LCALL → long call
- ② ACALL → absolute call

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{ LCALL vs ACALL } useful comparison for CT exam

RET 'Return Instruction'

P0	P1	P2	P3
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→ Micro controller registers ports

subroutine এ push করার পর pop না করলে কী হবে?

→ Subroutine will stuck.

LCALL 64KB

ACALL 2KB

ACALL can save a number of bytes of program ROM.

#page 121

important math problem

→ Clock cycle → Machine cycle | Definition

length machine cycle depends on the crystal oscillator of the chip
8051 microcontroller এ 12 oscillator periods.

$$T = \frac{1}{f}$$

#imp

$$\left(\begin{aligned} \text{freq} &= 11.0592 \text{ MHz} \rightarrow \text{অন্যক} \text{ freq} = 11.0592 / 12 = 921.6 \text{ KHz} \\ \text{machine cycle} &= 1 / 921.6 \text{ KHz} = 1.085 \mu\text{s} \end{aligned} \right)$$

→ এতটুকুই জন্য 1.5 Marks

②

page 122 describes which machine instruction take what time? (কৃত্রিম স্মৃতি হ'ল)

Find the delay of instruction of the program if crystal is 11.0592 MHz. #page 123

$$[(200 \times 2) + 1 + 2] \times 1.085 \mu s$$

#Page 124

Machine Cycle

DELAY :	MOV R3, #250	1
HERE :	NOP	1
	NOP	1
	NOP	1
	NOP	1
	DJNZ R3, HERE	2
	RET	2

$$(250 \times 4) + 1 + 2 = 1003 \times 1.085$$

$$= 1087.75 \mu s$$

Shayan

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200 [4x250]

page 126-127

Nested Loop (3 loop)

— page 128 —

#Microprocess

Chapter 1-4

ASCII

BCD

X Background

Number System ☒

Chapter 2/ page 72

AH	AL	64 bit → RAX	16-bit → AX
		RBX	→ BX
BH	BL	RCX	→ CX
		RDX	→ DX
CH	CL		
DH	DL	32 bit → EAX	
		EBX	
		ECX	
		EDX	

Multipurpose Register (Maruti's book)
Description

Flag register

PSW Register C, AC, OV, P

Zero flag = ~~result~~

if $Z = 1$ $res = 0$

if $Z = 0$ $res \neq 0$

Sign flag Left most bit = MSB

$S = 0/1$

Trap = Trap flag enables trapping
flag through an on-chip debugging

Interrupt flag =

$INTR = 1$ enable

$INTR = 0$ disable

D = Direction

1 = decrement

0 = increment

Z, S, T, I, D

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(DS, ES, SS) Maruti's books

X FS, AS

Real Mode vs Protected Mode

↓
1st 1MB
is real
memory
↓
DOS
Memory
System

Requirement of real mode

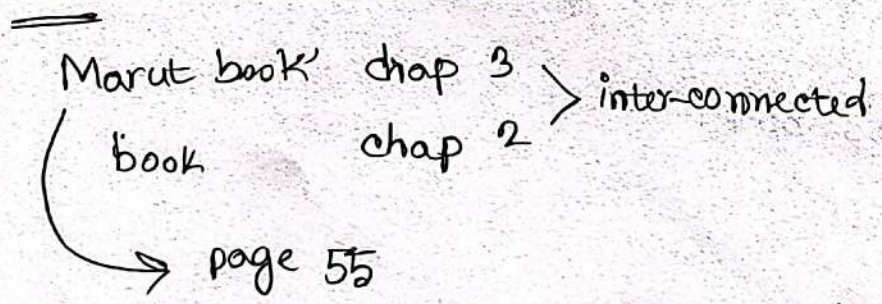
- ↳ Segment
- ↳ offset

Segment & offset = 7B page

Segment Register	Starting Address	Ending Address
2000H	20000H	2FFFFH
2001H	20010H	
2002H		
2100H	21000H	
AB00H	AB000H	
1234H	12340H	

Starting address = segment register x 10

Ending address = Starting + FFFF



Physical address = segment x 10 + offset (5bit)

logical address = segment : offset (4bit) ✓ Example 3.1

Example 3.2:

⑦

A memory location has physical address 80FD2H, in what segment does it have offset BFD2H?

$$80FD2H =$$